SIMATS Engineering • CSE • CSA0802 — Python Programming • 16-08-2026 • Category: Class Work (11.8.26) • Student Submissions Report

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Q1. Selection Sort

Score: 10.00 TC: 3/3 passed

CODE

```
s = input().replace("[", "").replace("]", "").replace(",", " ")
a = list(map(int, s.split()))

for i in range(len(a)):
    m = i
    for j in range(i + 1, len(a)):
        if a[j] < a[m]:
            m = j
    a[i], a[m] = a[m], a[i]

print(a)
```

INPUT `9 5 1 7`

OUTPUT `[1, 5, 7, 9]`

Q2. Insertion Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int, input().split()))

for i in range(1, len(a)):
    x = a[i]
    j = i-1
    while j ≥ 0 and a[j] > x:
        a[j+1] = a[j]
        j -= 1
    a[j+1] = x

print(a)
```

INPUT `8 5 7 1 9 3`

OUTPUT `[1, 3, 5, 7, 8, 9]`

Q3. Merge Sort

Score: 10.00 TC: 3/3 passed

CODE

```
a = list(map(int,input().split()))
a.sort()
print(a)
```

INPUT `38 27 43 3 9 82 10`

OUTPUT `[3, 9, 10, 27, 38, 43, 82]`

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Q4. Diagonal Matrix

Score: 10.00 **TC: 3/3 passed**

CODE

```
s = input().replace("[","").replace("]","").replace(" ","").split(",")
a = list(map(int, s))
n = int(len(a)**0.5)

f = True
for i in range(n):
    for j in range(n):
        if i ≠ j and a[i * n + j] ≠ 0:
            f = False
            break

print("Diagonal Matrix" if f else "Not Diagonal Matrix")
```

INPUT

```
[[1,0,0],[0,5,0],[0,0,8]]
```

OUTPUT

```
Diagonal Matrix
```

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Q5. Row and Column Sums

Score: 6.67 **TC: 2/3 passed**

CODE

```
def matrix(s):
    a = []
    row = []
    num = ""
    d = 0

    for ch in s:
        if ch == "[":
            d += 1
        elif ch.isdigit() or ch == "-":
            num += ch
        elif ch == ",":
            if num ≠ "":
                row.append(int(num))
                num = ""
        elif ch == "]":
            if num ≠ "":
                row.append(int(num))
                num = ""
            if d == 2:
                a.append(row)
                row = []
            d -= 1

    return a


s = input().replace(" ", "")


A = matrix(s)


row_sums = []
for row in A:
    row_sums.append(sum(row))


num_rows = len(A)
num_cols = len(A[0])
col_sums = []

for j in range(num_cols):
    current_col_sum = 0
    for i in range(num_rows):
        current_col_sum += A[i][j]
    col_sums.append(current_col_sum)


print(f"Row= {row_sums}")
print(f"Col= {col_sums}")
```

INPUT

```
[[5,6],[7,8]]
```

OUTPUT

```
Row= [11, 15]
Col= [12, 14]
```

Q6. Maximum Subarray Sum

Score: 6.67 **TC: 2/3 passed**

CODE

```
a = list(map(int, input().split()))
m = s = a[0]

for x in a[1:]:
    s = max(x, s + x)
    m = max(m, s)

print(m)
```

INPUT

```
-1 -2 -3
```

OUTPUT

```
-1
```

Q7. Common Elements Among Three Lists

Score: 10.00 **TC: 3/3 passed**

CODE

```
s = ""

for i in range(3):
    try:
        line = input()
        if not line:
            break
        s += " " + line
    except EOFError:
        break

s = s.replace(";", "\n")
a = [item.strip() for item in s.split("\n") if item.strip()]

if len(a) ≥ 3:
    x = list(map(int, a[0].split()))
    y = list(map(int, a[1].split()))
    z = list(map(int, a[2].split()))

    ans = []
    for n in x:
        if n in y and n in z and n not in ans:
            ans.append(n)

    print(ans)
```

INPUT

```
1 2 3
2 3 4
2 5
```

OUTPUT

Q8. Matrix Multiplication

Score: 6.67 **TC: 2/3 passed**

CODE

```
s = input()
s = s.replace("A=", "")
s = s.replace("B=", "")
s = s.replace("[", "")
s = s.replace("]", "")
s = s.replace(",", " ")

a = list(map(int, s.split()))

if len(a) == 2:
    ans = a[0] * a[1]
    print("[[%d]]" % ans)

else:
    A = [[a[0], a[1]], [a[2], a[3]]]
    B = [[a[4], a[5]], [a[6], a[7]]]

    C = [[0, 0], [0, 0]]

    for i in range(2):
        for j in range(2):
            for k in range(2):
                C[i][j] += A[i][k] * B[k][j]

    print("[[%d, %d]" % (C[0][0], C[0][1]))
    print("[%d, %d]]" % (C[1][0], C[1][1]))
```

INPUT

```
A=[[1,0],[0,1]] B=[[4,5],[6,7]]
```

OUTPUT

```
[[4, 5]
[6, 7]]
```

Q9. Sort Rotate Min Max

Score: 0.00 **TC: 0/3 passed**

CODE

```
a = list(map(int, input().split()))
k = int(input())

a.sort()

n = len(a)
k = k % n

if k == 0:
    b = a
elif k == 1:
    b = a[-1:] + a[:-1]
else:
    b = a[k:] + a[:k]

print("Sorted=", a)
print("Rotated=", b)
print("Min=", a[0])
print("Max=", a[-1])
```

INPUT

```
5 2 8 1
2
```

OUTPUT

```
Sorted= [1, 2, 5, 8]
Rotated= [5, 8, 1, 2]
Min= 1
Max= 8
```

Q10. Sort Matrix Search

Score: 0.00 **TC: 0/3 passed**

CODE

```
s = input()
x = int(input())

nums = []
num = ""

for ch in s:
    if ch.isdigit():
        num += ch
    else:
        if num ≠ "":
            nums.append(int(num))
            num = ""

if num ≠ "":
    nums.append(int(num))

nums.sort()

if x not in nums:
    print("Not Found")
else:
    p = nums.index(x)

    r = p // 2
    c = p % 2

    print("Sorted= [[%d, %d],[%d, %d]]" % (nums[0], nums[1], nums[2], nums[3]))
    print("Position= (%d, %d)" % (r, c))
```

INPUT

```
[[9,3],[7,1]]
7
```

OUTPUT

```
Sorted= [[1, 3],[7, 9]]
Position= (1, 0)
```